

Automatic Speech-to-Speech Translation of Educational Videos Using SeamlessM4T and Its Use for Future VR Applications

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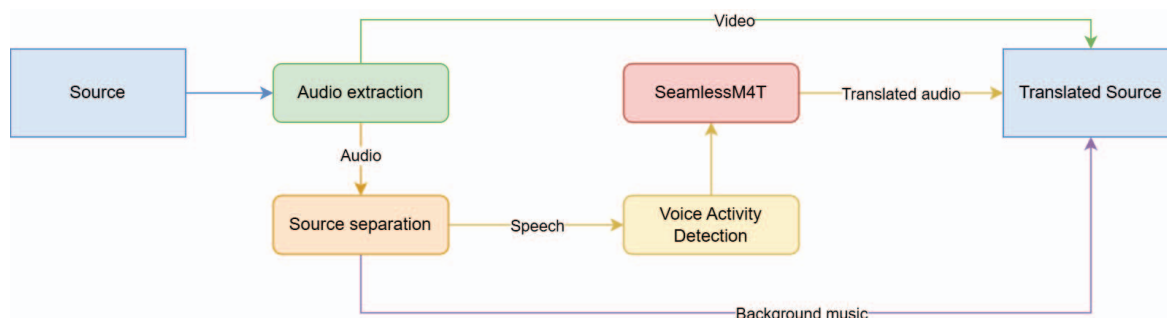


Figure 1: Main pipeline used to automatically translate educational content. Besides the use of the end-to-end Seamless model, we also used additional models to extract and segment the speech content before providing it to the translation model.

ABSTRACT

Automatic Speech-to-Speech Translation (S2ST) is crucial for VR, providing immersive experiences and global accessibility. For this task, cascade pipelines are often used, but it faces challenges in low-resource languages due to data scarcity, complexity, and maintenance, meanwhile end-to-end models, though promising, are still in early development. This study explores the latest SeamlessM4T model, an end-to-end S2ST architecture showing great potential for VR applications, and discusses its strengths and limitations in the context of educational VR for low-resource languages.

Index Terms: Speech-to-Speech Translation—Speech Translation—;—Low-resource languages—

1 INTRODUCTION

Machine learning (ML) and artificial intelligence (AI) have played a crucial role in advancing speech technologies in recent years [8, 16]. Some of these technologies include Automatic Speech Recognition (ASR) [1, 3, 7, 10], Text-to-Speech (TTS) [19], Speaker Diarization and Identification [4], among others.

Despite the massive progress, the development of speech technologies are still absent to the majority of spoken languages around the word [9]. These technologies, including the task of Speech-to-Speech Translation (S2ST), usually requires large amounts of data and resources to train the models [1]. In contrast, more resource languages such as English have the last of speech technologies available. In this context, speech technologies for other languages might help to address the linguistic inequality and digital divide faced by communities speaking less widely spoken languages. In special, S2ST technologies not only could help to provide content of a diversity of languages for those communities, but also enabling these

communities in sharing their own knowledge with all the people around the word [2].

Differently than Machine Translation (MT), that is a task where written text in one language is translated into another [5, 14], Speech-to-Speech Translation involves the direct translation of spoken language. In this task, both the input and output are in the form of speech instead of written text.

This distinction brings both advantages and disadvantages. On the positive side, Speech-to-Speech Translation provides a more natural and immediate mode of communication, preserving intonation, emotion, and other nuances present in spoken language. It is particularly useful in scenarios where seamless communication is crucial, such as during real-time conversations. However, besides the complexity of the pipeline, the computational cost of processing speech in real-time poses technical challenges, making Speech-to-Speech Translation a task that requires robust solutions to ensure both accuracy and efficiency.

By enabling S2ST seamlessly, individuals using low-resource languages can access and contribute to Virtual Reality (VR) applications, particularly in educational contexts, where the impact of such technology can be transformative. In educational VR contexts, the impact of seamless Speech-to-Speech Translation (S2ST) is multifaceted. Firstly, it fosters an inclusive learning environment by breaking down language barriers, allowing students and educators using low-resource languages to participate fully in immersive educational experiences. Secondly, S2ST promotes cross-cultural collaboration, facilitating communication and knowledge exchange among students and educators from diverse linguistic backgrounds. This not only enriches the educational experience but also cultivates a global perspective among learners. Additionally, the technology enhances the accessibility of educational content by providing real-time translation during virtual lectures, discussions, or interactive simulations, ensuring that information is conveyed effectively to all participants. As a result, seamless S2ST not only can potentially democratize the access to educational VR applications but also contribute to the creation of a more interconnected and inclusive global learning community. Fortunately, recent advances have shown promising results in these scenarios [2].

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The development of speech translation solutions poses significant challenges for the following reasons:

- Lack of data in low-resource languages: the majority of languages lack sufficient datasets for training models dedicated to speech translation. This scarcity hinders the ability to develop robust solutions for these languages.
- Complexity of Multimodal Data and lack of parallel data: speech translation encompasses various sub-tasks, including machine translation, speech recognition, and speech synthesis. Furthermore, obtaining parallel data for training speech-to-speech models directly is notably challenging. The scarcity of such aligned datasets complicates the training process.
- Highly complex pipelines: while cascade pipelines are generally easier to implement compared to end-to-end models, their ability to generalize across diverse domains and languages often necessitates retraining. This introduces an additional layer of complexity in the development process.
- Seamless translation (real-time): the complexity of speech translation models requires substantial computational resources, which can impede real-time performance. The demand for computational power may hinder the seamless translation capabilities of these systems, particularly when efficiency in real-time processing is required;
- Prosody and timbre maintenance: Many Speech-to-Speech Translation (S2ST) systems often overlook the crucial aspects of prosody and timbre preservation during the translation process [13]. As the demand for natural and human-like speech in translation grows, the significance of retaining the original speaker's intonation, rhythm, and tone becomes increasingly apparent. The lack of attention to prosody and timbre can result in translated speech that sounds robotic or unnatural, diminishing the overall quality and effectiveness of the translation. Addressing these factors is essential for achieving more authentic and contextually appropriate speech translations, contributing to a more immersive and user-friendly communication experience.

This work is organized as follows: Section 2 explains the task in detail. Section 2.1 describes the SeamlessM4T model. Section 3 describes the methods used for the demonstration. Section 4 discusses the results. Finally, Section 5 concludes the paper.

2 SPEECH-TO-SPEECH TRANSLATION

Speech-to-Speech Translation (S2ST) is a task that involves the conversion of spoken language in one language (source language) into spoken language in another language (target language). There are two main strategies for S2ST systems: cascade and end-to-end. In the cascade approach, the S2ST system is divided into distinct modules or stages, each dedicated to specific tasks such as Automatic Speech Recognition (ASR), Machine Translation (MT), and Text-to-Speech (TTS). The output of one module serves as the input for the next, creating a sequential pipeline. This strategy allows for a modular and interpretable system design, where individual components can be optimized independently. However, it may face challenges in handling end-to-end optimization and capturing complex dependencies across different tasks. In the end-to-end approach, such as Seamless M4T [2] and StyleS2ST [13], the systems aim to integrate the entire S2ST system into a single, unified model. This model directly translates input speech in the source language to output speech in the target language without explicitly separating tasks such as ASR, MT, and TTS. End-to-end models have the advantage of potentially capturing more nuanced interactions between linguistic elements and providing a simpler and more streamlined

architecture. However, training end-to-end models can be computationally demanding, and they may require large amounts of data to generalize effectively across diverse scenarios.

Recently, the development of direct speech-to-speech translation (S2ST) systems has gained prominence, driven by advances in speech processing architectures. Among the recent models that stand out in this scenario, we can mention SeamlessM4T [2], StyleS2ST [13], and DASpeech [6].

StyleS2ST [13] is a model that stands out for its unique ability to transfer speech styles from a source language to a target language. While previous studies focused primarily on the accuracy of semantic translation, this model addresses the transfer of speech styles, such as timbre, intonation, and rhythm. To overcome challenges such as the scarcity of high-quality parallel data, researchers built a parallel corpus using a multilingual and multi-speaker speech synthesis system, resulting in the proposed StyleS2ST.

DASpeech [6] is another highlight, presenting itself as a non-autoregressive two-step model for S2ST. Its innovative approach utilizes a two-step-based architecture, aiming to overcome challenges related to linguistic and acoustic diversity. Its strategy provides significant improvements in translation quality and decoding efficiency.

Finally, SeamlessM4T [2] is a model that seeks to provide efficient and unified translation between speech and text in up to 100 languages. While text translation models have advanced for over 200 languages, speech-to-speech translation still faced challenges. The training of this model involved the use of 1 million hours of speech audio data, resulting in significant improvements compared to existing models.

2.1 SeamlessM4T

SeamlessM4T v2 represents the second iteration of the SeamlessM4T model, retaining all the capabilities of its predecessor while introducing additional features. Notable enhancements include low-latency support and the preservation of prosody. This upgraded model excels in translating content from 100 source languages to over 36 target languages, delivering the output in the form of speech. Beyond Speech-to-Speech Translation (S2ST), SeamlessM4T v2 extends its functionality to encompass tasks such as automatic speech recognition and speech-to-text translation [2]. In contrast, the SeamlessExpressive version, designed specifically for preserving prosodic and timbre characteristics, is optimized for translation to a more focused set of languages. Currently, SeamlessExpressive supports translation into six languages: English, French, German, Italian, Mandarin, and Spanish. This version is tailored to cater to users seeking the nuanced preservation of expressive elements in their translated speech, ensuring a natural and authentic output in a select set of languages.

The architecture of the model is depicted in Figure 2. The speech translation task is deconstructed into two distinct subtasks: speech-to-text translation (S2TT) and text-to-unit conversion (T2U). This high-complexity model involves encoding both the speech and text inputs into units. It further predicts the duration of these units in the output and generates a new waveform through the utilization of a unit-vocoder. This process enables the model to effectively transform speech inputs into corresponding textual representations, predict the duration of the units in the output and generates a new waveform using a unit-vocoder [2].

3 DEMONSTRATION

Figure 1 illustrates the comprehensive pipeline employed in our system. In addition to leveraging the end-to-end Seamless model, we incorporated supplementary models to extract and segment the speech content before presenting it to the translation model. To showcase the system's capabilities, we curated a selection of videos

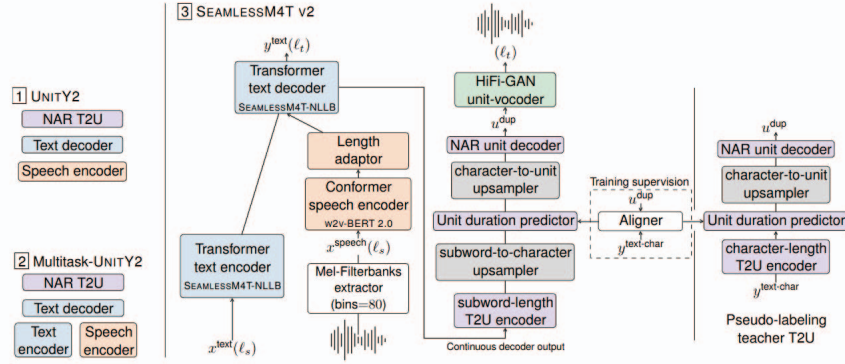


Figure 2: Architecture of SeamlessV2 M4T model [2].

from TedED, a branch of TED¹, specifically focused on education by providing a platform for educators to create and share lessons in the form of animated videos.

Our evaluation focused on translating content from English to two other languages: Spanish and Portuguese. In Spanish we used the SeamlessExpressive version of the model to emphasize prosody and to preserve the timbre in the translated output. This emphasis on maintaining the original speaker’s intonation and tone is vital for ensuring a natural and high-quality translation experience, but unfortunately it is not available for all the languages.

For the video and audio separation, we used Demucs [12], a state-of-the-art music source separation model based on a U-Net convolutional architecture inspired by Wave-U-Net [15] and Transformers [18]. We extracted two stems from the audio source, vocals and non vocals, and used the vocals part for the next step of the pipeline.

Voice Activity Detection (VAD) is a technology utilized in speech processing to discern segments of an audio signal containing human speech and differentiate them from non-speech segments. In our implementation, we employed the Silero VAD [17], a model recognized for its robust performance across various scenarios. For the demonstration, we also saved the timestamps where each segment was present in the original audio to reconstruct the video with the translated audio in the end of the process. In our case, this step is essential for handling small audio segments and reconstructing the translated video with accurate timestamps. One can notice that, while VAD may not be mandatory for real-time conversations, it can prevent the unnecessary consumption of computational resources.

Following the segmentation process, we proceeded to translate the audio segments using the SeamlessM4T model. For English to Spanish translation, we utilized SeamlessExpressive, a model specifically designed to automatically preserve the timbre and prosody of the original audio. This choice ensures that the translated content maintains the distinctive qualities of the original speaker’s voice, contributing to a more natural and expressive output.

For English to Portuguese translation, we employed SeamlessM4T v2 Large. It’s important to note that, in this particular instance, the timbre underwent a transition from a female to a male voice, a characteristic beyond our control. Additionally, the prosody also changed, resulting in a translated version that aligns with a readable speech of the target language.

4 DISCUSSION

This section discusses the result using our proposed pipeline. Initially, we examine the translation performance obtained in our ex-

periments, followed by a discussion of the potential applications for VR.

4.1 Performance analysis

The utilization of Seamless models in our speech translation system presents several advantages over alternative technologies, contributing to the overall effectiveness and user experience. One notable advantage is the model’s capacity to seamlessly preserve the original content, timbre, and expressivity of the source audio during translation. This characteristic is particularly crucial for maintaining the naturalness and authenticity of the speaker’s voice, enhancing the overall quality of the translated content.

Comparing different Seamless models, we observed that SeamlessExpressive, specifically designed for voice preservation, excelled in maintaining the timbre and prosody of the original speech during English to Spanish translation. This resulted in a more expressive and contextually relevant output, contributing to a higher level of user engagement and comprehension. On the other hand, SeamlessM4T v2 Large, is a more generalist model, with better support for different languages. In this scenario, there are still room for research in the topic regarding the generalization of SeamlessExpressive and similar technologies across different languages, specially the low-resource ones.

An example of the transcription for a segment of our demo is provided in Table 1. We observed a decline in translation performance into Portuguese during the demonstration. Some segments were left untranslated, despite a change in the speaker’s timbre. In these instances, the speaker appeared to switch between Portuguese and English for certain words. Other effect observed was the presence of hallucination, a common problem in large language models (LLMs) [11]. The cause of these outcomes are unclear, whether it stems from a generalization issue in the model or if it is specific to the particular vocabulary of the video utilized in this case. In the case of Spanish, we noted improved accuracy, with only a few sections remaining untranslated or containing errors.

4.2 Seamless Models for Educational VR

While Seamless models excel in enhancing the overall user experience in speech translation, their application extends beyond conventional scenarios into emerging technologies such as Virtual Reality (VR) and education. In the context of VR, Seamless technology can contribute to a more immersive and natural experience by providing real-time, language-specific voice translations within virtual environments. This capability not only facilitates seamless communication among users speaking different languages in a virtual space but also enhances the authenticity of VR interactions.

¹<https://www.ted.com/>

Table 1: Example of transcribing an excerpt and the expected translation. The original sentence is “*For example, a 2017 study found that those who were highly focused on the outcomes of their New Year’s resolutions, or driven by extrinsic motivation, weren’t the most likely to stick to them.*”. Errors are highlighted for clarity.

Language	Expected	Transcription
Spanish	Por ejemplo, un estudio de 2017 encontró que aquellos que estaban muy centrados en los resultados de sus propósitos de Año Nuevo o impulsados por motivación extrínseca no eran los más propensos a cumplir con ellos.	Por ejemplo, un estudio de 2017 encontró que aquellos que estaban muy enfocados en los resultados de sus resoluciones de Año Nuevo o impulsado por ECEX.
Portuguese	Por exemplo, um estudo de 2017 descobriu que aqueles que estavam altamente focados nos resultados de suas resoluções de Ano Novo, ou impulsionados por motivação extrínseca, não eram os mais propensos a cumpri-las.	Por exemplo, um estudo de 27 anos descobriu que aqueles que são altamente focados em the outcomes of the New Year’s Resolutions.

In education, the integration of Seamless models can revolutionize language learning and accessibility. By offering real-time translation of educational content, Seamless ensures that language is no longer a barrier for students accessing lessons in their native language. Moreover, in VR-based educational experiences, Seamless technology can create a dynamic multilingual environment, allowing students from diverse linguistic backgrounds to collaborate and engage in interactive learning activities.

Another example is the potential application of Seamless models in overcoming language and geographical barriers for agile teams in a VR environment. Mentors and students from different countries can collaborate seamlessly, leveraging real-time, language-specific voice translations provided by Seamless technology. Simultaneous translation streamlines communication, fostering a more inclusive and effective collaborative atmosphere, showcasing the transformative impact of Seamless models in diverse global contexts.

Besides the promising results, there is still room for research, particularly in the generalization of SeamlessExpressive and similar technologies across different languages, especially those considered low-resource. Investigating the adaptability and effectiveness of these models in preserving voice characteristics in languages with limited training data is crucial for expanding their applicability and impact.

5 CONCLUSION

Automatic speech translation in VR technologies promotes inclusivity, accessibility, and realistic communication, making virtual experiences more enjoyable and effective for a diverse global audience. The integration of Seamless models in VR technologies and educational settings holds tremendous promise. In this work, we developed a demo for automatically transcription of educational content using the state-of-the-art of speech-to-speech multilingual translation. The results are promising even for low-resource languages. The preservation of voice characteristics, real-time translation capabilities, and adaptability to diverse linguistic scenarios position Seamless as a valuable tool for creating inclusive and immersive experiences in the realms of virtual reality and education. Ongoing research in this field will likely lead to further advancements, making these technologies even more accessible and effective for a global audience.

As future works, we highlight the importance of fine-tuning the

SeamlessExpressive model for other languages. As the preservation of prosody and voice timbre are crucial for maintaining the naturalness of speech, extending the capabilities of the SeamlessExpressive model to encompass a broader range of languages is a key avenue for future research.

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